

Brazil

Administrative Council for Economic Defense (CADE)

I. Context

The Brazilian Competition Authority has considerable experience in the development of data mining and screening techniques to enhance the detection of signs of bid-rigging. The Cerebro project harnesses a wide array of computational tools to detect and investigate cartels operating in public procurement processes, capitalizing on Brazil's robust culture of public data accessibility.

One example of this culture are public datasets: Comprasnet, the Brazilian Federal Government's Procurement Portal launched in 1997, electronically publishes bidding notices, calls for bids (of all modalities), and summaries of contracts signed by the Federal Public Administration. This movement towards transparency continuously grew over the last years and recently was enforced by the enactment of a new public procurement law in 2021. The new public procurement law mandated that all Brazilian procurements be online (with few exceptions). Subsequently and related to Brazilian public data culture, the Portal Nacional de Compras Públicas (PNCP) was set up to spread comprehensive information about tenders, bidders, bids, winners, reference prices, and related data, including all states and municipalities.

Given this extensive range of public procurement data, the first challenge lies in accessing and processing these datasets to facilitate Cerebro's primary aim: the detection of collusive behavior. In this context, the data must be selected and processed as close as possible to the needs of Cade's anticartel core business, which is intrinsically related to the active search of new cases on one hand, and the enrichment of opened cases with relevant data on the other hand. This work is done alongside other units of the General Superintendence.

II. Techniques and Computational Tools

A significant challenge for Cerebro is developing methodologies and tools that empower case handlers in identifying companies engaged in illicit activities during ongoing investigations. Also, there is a need to help those who are responsible for handling complaints. In this regard, Cerebro's products can speed up responses to public claims.

The first generation is built upon techniques like network analytics, statistical, and econometric indicators. Networks elucidate firms' societal compositions, potential shared decision boards, and the organization of tenders and bidders. Traditional indicators, such as variance, cover bidding and structural screens, are currently implemented as Business Intelligence (BI) panels.

The next generation of tools is under development, harnessing modern machine learning techniques, including artificial intelligence (AI) and open-source intelligence. Ongoing research investigates unsupervised learning to address the challenge of limited labeled data. Clusters are defined using various inputs, including bid-rigging screens and communication evidence. Additionally, Cerebro plans to employ natural language processing (NLP) and large language models (LLMs) to enhance the capabilities of procurement document processing, particularly in the search for documents' coincidences and other indirect communication evidence. All these datasets and information can be used as input data to the machine learning model.

To make all the tools mentioned available for the investigators, Cerebro's team must have the ability to build them from the ground up performing:

- **Requirement elicitation:** includes not only talking to case handlers, but also managing cases. The General Superintendence decided on a development tailored to the challenges faced by the investigators.
- **Software development:** Backend (including data security, database operations, API development and dev-ops) and Frontend (User Interface (UI) and User Experience (UX) design).

It is worth mentioning that previously to the implementation of a tool for the investigator's direct use, like a web-app, there is all the research and development that is made using the mainstream data-science toolkit: *jupyter notebooks*, quarto-markdown documents, R and Python scripts, and so on. At the same time, Cerebro's team is also clearly tailored to innovating not only on the techniques itself. So, there are PoCs (proof of concepts) and prototypes of solutions incorporating more performant and secure languages like Rust into the data science workflow.

Cerebro's team is also currently developing a platform, which is a web-app to host the most used tools by the investigators, using the newest architecture and web technologies for the backend, like Docker, Kubernetes, Microservices and Go, alongside more established technologies like Angular for the frontend. The aim is to support the investigators and case handlers from Cade's General Superintendence.

The methodologies and tools aid in identifying companies engaged in illicit activities during ongoing investigations. As a result, it is crucial to propose indicators or markers that simplify the detection of patterns typically associated with collusive behavior.

The information employed by Cerebro in antitrust investigations can be categorized as follows:

1. Information focused on firms: Encompasses details about companies and their systems, including office locations and the IP addresses used during participation in electronic tenders, for example.
2. Information focused on tenders: Focuses on the behavior of firms within tenders, analyzing participation profiles, including firms frequently taking part together, firms winning tenders but not fulfilling contracts ("rabbits"), firms consistently unsuccessful in tenders, and firms exclusively winning in specific regions.

Both categories are complementary and instrumental in comprehending market behavior. The former finds suspicious coincidences, like shared IP addresses, potentially indicating illicit communication. The latter, coupled with traditional economic screens like irrelevant bids, contributes to the body of evidence suggesting the existence of a cartel.

It is important to note that, as most antitrust agencies, Cerebro lacks access to a comprehensive database with labeled data clearly identifying collusive firms and their corresponding markers. Given this constraint, Cerebro focuses on providing comprehensive information to investigators, who have the ability to discern their relevance within specific case contexts.

The conversion of data into useful information in the context of a case is not straightforward, particularly when there are multiple data and pieces of information that come from other sources. Consequently, Cerebro develops tools that streamline the interpretation of results for non-data scientist experts, facilitating informed decision-making in cases. To this end, Cerebro's team is actively curating a catalog of tools.

These tools can be divided into two generations.

III. Challenges

Despite all the technical and methodological advances achieved in the last two years, Cade still faces some challenges to reaching its full potential that are worth mentioning. First, the reports produced by Cerebro analysis must be subjected to the scrutiny of professionals who have low or no knowledge about the technologies used, economics, and statistics theory. For example, the judges responsible for authorizing dawn raids could face difficulty understanding and interpreting the data produced. To avoid misinterpretation is necessary for the Cerebro team to clearly communicate the methods, techniques, and results in such a way that is understandable by the average law professional.

Another concern is about the Cerebro project staff. It is necessary to request professionals from other agencies, which is hard, given the scarcity of qualified professionals with the necessary skills in public service. Consequently, selecting and keeping this kind of skilled professional is a great challenge that Cade faces. Furthermore, not only technical skills are needed. The professional must be an analytical person by nature and have soft skills like curiosity and openness to collaborate and engage in interdisciplinary discussions. Those hard and soft skills can be developed, but there is a steep learning curve.

Regarding the demobilization of the data science team risk, Cade encourages employees to participate in long-term training plans and constantly register and manage knowledge of all products developed in the project.

In addition, strengthening partnerships with other Brazilian investigative bodies and other countries' antitrust agencies is also a strategy to face those challenges. The experience of listening and learning from others is enriching, that's why the Cerebro Project is always open for international collaboration, like in the recently established ICN Technologist Group Work.